

Finger & Hand Structure for Climbers

Core Learning · Foundations of Human Anatomy & Physiology

How the fingers are built, how they bend, and how load actually travels through them when climbing.

Climbing places more demand on the fingers than almost any other sport. Yet very few coaches or athletes have a clear picture of what is actually happening inside a finger when it's loaded on a small edge. This lesson builds that picture — not at a medical or clinical level, but at the level a coach needs: enough to understand why certain grip positions and hold sizes carry more risk than others, and why training and injury management decisions are built on this anatomy in the first place.

1. The Bones and Joints of the Wrist, Thumb & Fingers

The Wrist

The wrist is not a single joint but a complex of eight small bones called the carpal bones, arranged in two rows between the forearm and the hand. The proximal row consists of the scaphoid, lunate, triquetrum, and pisiform. The distal row consists of the trapezium, trapezoid, capitate, and hamate. These bones allow the wrist to flex, extend, deviate side to side, and rotate in a limited arc.

For climbers, the most clinically relevant carpal bone is the scaphoid, vulnerable to injury in hard landings. The hamate's hook, on the small-finger side, is a known stress site in climbers who repeatedly load the pinky-side of the hand in specific grip positions.

Another structure worth knowing is the **triangular fibrocartilage complex (TFCC)** — a disc of cartilage on the small-finger side of the wrist, acting as a cushion and stabiliser. In climbing, the TFCC is most commonly stressed during sidepull and undercling moves, and any position loading the wrist while twisted toward the small-finger side (ulnar deviation).

The wrist joint complex forms the carpal tunnel — a fibrous roof (flexor retinaculum) holding the long flexor tendons close to the wrist as they pass through to the fingers. Wrist position during a grip directly affects tendon mechanical advantage — a flexed wrist reduces grip strength significantly, which is why coaches cue a neutral or slightly extended wrist on the wall.

The Thumb

The thumb differs from the fingers in two ways: it has only two phalanges rather than three, and its base joint — the carpometacarpal (CMC) joint — is a saddle joint rather than a hinge, giving it the ability to move across the palm (opposition), essential for pinch grips, side-pulls, and underclings.

- **CMC (carpometacarpal joint)** — the base joint at the wrist. A saddle joint allowing multi-directional movement. Most commonly affected by overuse in climbers who train pinch grip heavily.
- **MCP (metacarpophalangeal joint)** — the large knuckle of the thumb. Stabilised by the ulnar collateral ligament (UCL) — the structure injured in 'skier's thumb'/gamekeeper's thumb.
- **IP (interphalangeal joint)** — the single middle joint of the thumb, equivalent to the PIP of a finger. Involved in pinch and compression moves.

The Fingers

Each finger — excluding the thumb — is built from three phalanges (proximal, middle, distal), connected by two main hinge joints plus a third joint at their base, making three load-bearing joints per finger.

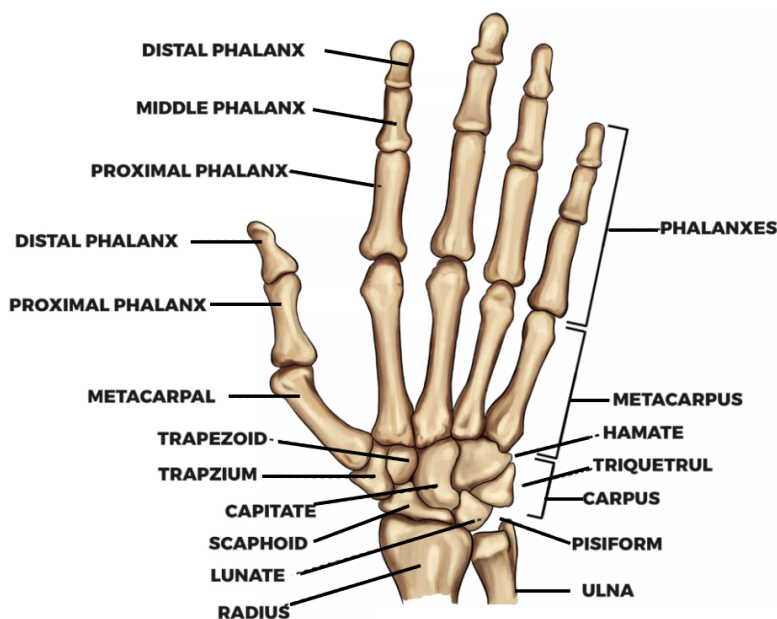
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- **MCP** — the large knuckle where finger meets hand. Flexes during all grip types; small side-to-side spread relevant in wide pinches and palming.
- **PIP (proximal interphalangeal joint)** — the middle joint; the largest and most powerful. Bends most during a crimp grip. Supported by the volar plate, which resists hyperextension. The most commonly injured joint in climbing, primarily through pulley strain and volar plate sprains. The A2 pulley sits just proximal to this joint.
- **DIP (distal interphalangeal joint)** — the small joint closest to the fingertip. Bends most during open-hand/drag grip. The A4 pulley crosses this region.

The amount each joint bends changes with grip type, and that difference matters. A tighter PIP bend (crimping) concentrates force through the A2 pulley and the FDS tendon. A deeper DIP curl (open-hand/drag) shifts demand toward the FDP tendon and the A4 pulley.

Coaches should think of grip type as a decision about where load gets concentrated, not simply a stronger or weaker hand position. Both crimping and open-hand positions carry injury risk — they just load different structures.



The bones of the hand and wrist

2. The Muscles of the Hand and Wrist

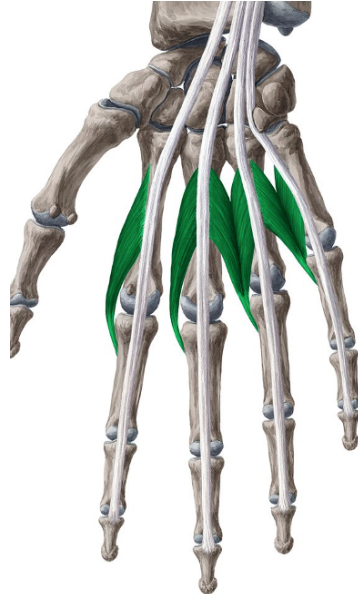
A common misconception is that finger strength comes from the hand itself. In fact, the primary muscles responsible for finger flexion — the flexor digitorum superficialis and flexor digitorum profundus — sit in the forearm, transmitting force to the fingers via long tendons through the carpal tunnel. The wrist flexors and extensors are also forearm muscles, which is why forearm fatigue and pump are the dominant limiting factor in climbing endurance.

The hand's own intrinsic muscles are largely about stability and fine motor control rather than raw force. The lumbricals coordinate MCP flexion with IP extension — subtle but important when transitioning between grip positions. The interossei muscles spread and draw the fingers together, contributing to precise finger placement.

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The thenar and hypothenar muscle groups control thumb opposition and small-finger positioning respectively. Training grip strength is fundamentally forearm training, while hand stability and precision develop through movement practice on the wall.



The lumbrical muscles — intrinsic hand muscles coordinating MCP flexion with IP extension

3. The Flexor Tendons — Where Finger Strength Actually Comes From

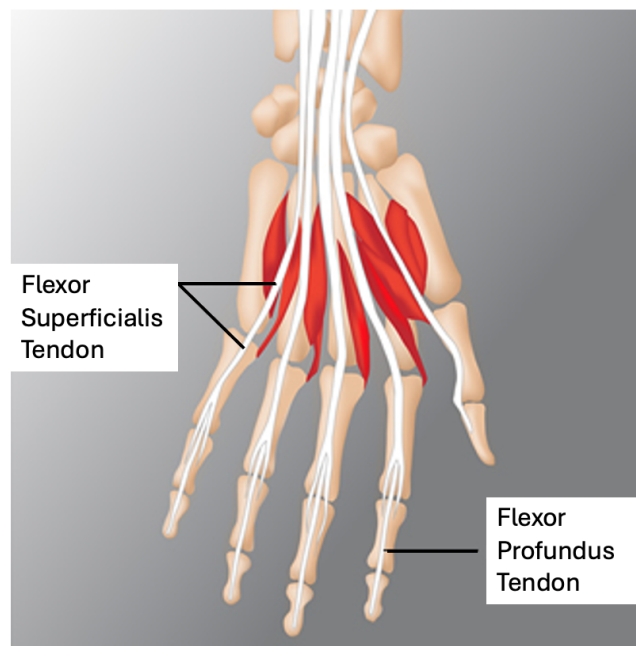
A detail that surprises many climbers: there is no muscle tissue inside the fingers themselves. Every bit of finger-closing force is generated by muscles in the forearm, transmitted down through long tendons.

- **FDP (flexor digitorum profundus)** — runs the full length of the finger, attaches at the fingertip bone. Generates the majority of pulling force; the primary tendon loaded when hanging on small edges.
- **FDS (flexor digitorum superficialis)** — attaches partway down the finger, near the PIP joint. Plays more of a stabilising role, sharing and distributing load.

Both tendons are under significant tension during almost any climbing grip, but the FDP is subjected to very high force when hanging on small holds — exactly why finger and forearm training protocols load this tendon system progressively rather than all at once.

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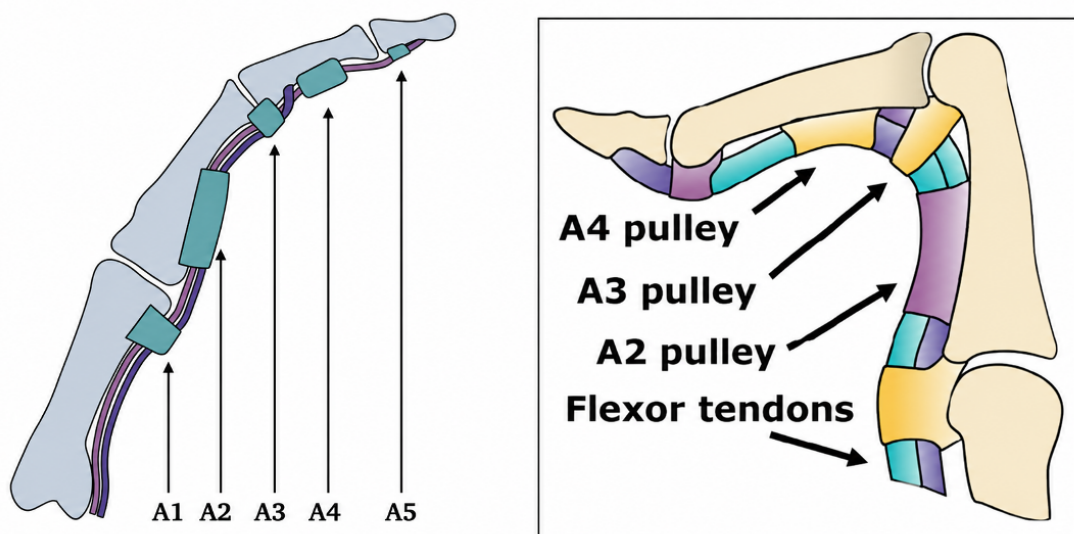
The flexor digitorum superficialis and profundus tendons

4. The Pulley System — Why This Is Where Most Finger Injuries Happen

If the flexor tendons are the cables that pull the finger closed, the pulleys are the structures keeping those cables running close against the bone rather than lifting away from it — small, tough rings of connective tissue wrapped around the tendon at intervals down the finger.

Without a functioning pulley system, a tendon under load would lift away from the bone in a straight line — bowstringing. Healthy pulleys prevent this by holding the tendon tight against the finger's natural curve.

- **A2 pulley** — near the base of the finger. The single most commonly injured pulley in climbing, experiencing the highest mechanical stress during crimped grip positions.
- **A3 pulley** — a smaller pulley close to the PIP joint.
- **A4 pulley** — located roughly midway down the finger.



The annular pulley system, A1-A5

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The A2 pulley takes the brunt of climbing injuries because of grip mechanics: a tightly flexed PIP joint (crimping) dramatically increases the angle — and therefore the force — the tendon exerts against the A2 pulley as it changes direction. This is a mechanical reality of the crimp position, not a matter of an individual climber's strength or technique being inadequate.

COACHING TAKEAWAY

Understanding the pulley system explains why a full crimp grip carries meaningfully more structural risk than an open-hand grip on the same hold — the difference is not just about which grip 'feels stronger', but about how much force each grip routes through the A2 pulley specifically.

5. How Load Actually Travels Through the Finger

Putting the previous sections together gives a coach a genuinely useful model: when a climber weights a hold, force travels through a chain of structures, each experiencing a different share of that load depending on grip position and hold size.

In simplified form, the chain runs: hold → fingertip (DIP) → middle joint (PIP) → A2 pulley → flexor tendon → forearm muscle.

- Smaller edges increase tendon and pulley force for the same body weight, since less surface area distributes the same load.
- A deeper PIP bend (crimping) increases stress specifically at the A2 pulley.
- A deeper DIP curl increases demand on the FDP tendon specifically.
- An unstable or inconsistent joint angle makes force distribution unpredictable repetition to repetition — part of why controlled, repeatable positions matter in structured training.

This model is also exactly why sensible finger training progresses gradually. Tendons and pulleys adapt to load through slow structural changes such as increased collagen density, far more slowly than muscular strength gains. A climber can become stronger in their forearm muscles well before their pulleys and tendons have caught up — precisely the mismatch that leads to injury when hold size or intensity increases too quickly.

Summary

A climbing finger is a simple but genuinely high-stress structure: three bones, two major bending joints, two flexor tendons doing all of the pulling work, and a series of pulleys holding those tendons in place. Grip type and hold size directly determine how load is distributed across this system — and because the pulleys and tendons adapt to load far more slowly than muscle does, this anatomy is the foundation for every sensible decision a coach makes about grip progression, hold selection, and training load.

FURTHER READING

Max Climbing — "Finger Structure for Climbers: Bones, Tendons & Pulleys (Simple Overview)"

maxclimbing.com/blogs/knowledge-hub-anatomy/finger-structure-for-climbers-bones-tendons-pulleys-simple-overview

TrainingBeta — Dr. Tyler Nelson, "The Simplest Finger Training Program"

trainingbeta.com/the-simplest-finger-training-program

End of Lesson 04. Return to corelearning.co.za to complete the Lesson 04 quiz (80% pass mark) and unlock the next lesson.